Lecture 10:

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Slides adapted from Tony Liu
And inspired by
https://web.stanford.edu/class/datasci112/lectures/lecture18.pdf

Logistics

- HW4 due this morning
- HW5 will be released sometime this week
  - No content from this lecture on it, I think, so it doesn't make sense to start it before next week's lecture anyway

Week 10

- HTML Fundamentals in order to build up to scraping
- Web scraping
- BeautifulSoup
- Containerization

DISCLAIMER: This is not an HTML class

What this means…

- It’s not my specialty
- Being able to code in HTML is not expected, just parsing common tags, like but not limited to:
  - table, tr, td, th
  - p
  - l

HTML Fundamentals

- Hyper Text Markup Language: standard markup language for creating webpages
- HTML documents contain a series of elements defined by tags (indicated by <>) that tell browsers how to display content
  - (example in 2 slides)
- HTML elements may have additional attributes specified within the tags
- See Mozilla web docs as a reference

Typical web technology stack

- HTML
  - structures the contents of a web page
- CSS (Cascading Style Sheets)
  - visually style the HTML content
    - We will take a look at scraping CSS very slightly today, but afaik you don’t need to understand it in order to use the BeautifulSoup methods, just HTML for our purposes
- JavaScript
  - provides dynamic scripting of web page functionality

Anatomy of an HTML element

Opening tag

Closing tag

```
<p class=”nice”>Hello world!</p>
```

An attribute and its value

Enclosed text content

An element is “void” if they do not have a closing tag

Programming analogy

- elements: objects of differing types, defined by tags
- attributes: instance attributes, not displayed to the end user
    - class: loosely like Python classes, allow CSS and Javascript to access specific elements with the same class
    - id: identifier specific to an element that is unique to the entire HTML document

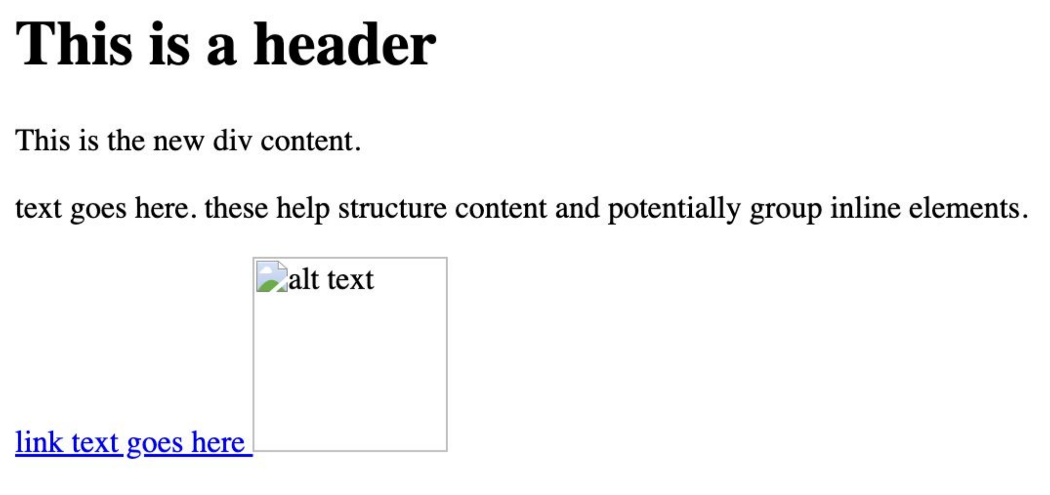
HTML Cheat Sheet of common elements

lecture10.html > ...

```html
<h1> This is a header </h1>
<p>
    text goes here. these help structure
    content and potentially group inline
    elements.
</p>
<a href="url"> link text goes here </a>
<img src="img_src" alt="alt text" width="100" height="100"
/>
```

Links →

- a for anchor element
- `href=` contains the URL



Div element

- content division element <div> </div>: does not inherently represent anything
  - used to group elements to be styled or manipulated by CSS/JS
- structurally, all webpages have a hierarchical tree-like nesting of elements, often organized by divs
  - allows for programmatic traversal scraping

```
<div>
    <p>This is the new div content.</p>
</div>
```

Tables

- <table>
  - element creates a table
- <tr>
  - is a row
- <th> and <td>
  - create a cell within a row

```
<table>
    <tr>
        <th>Position</th>
        <th>Author</th>
        <th>Books Published</th>
    </tr>
    <tr>
        <td>1</td>
        <td>Jane Austen</td>
        <td>6</td>
    </tr>
    <tr>
        <td>2</td>
        <td>George Orwell</td>
        <td>9</td>
    </tr>
</table>
```

Position	Author	Books Published
1	Jane Austen	6
2	George Orwell	9

Dynamic vs static webpages

Source

- Static websites
    - Deliver content the same for every visitor
    - Ex: Wikipedia
- Dynamic websites
    - most modern websites
    - Generate content in real-time based on user input, location, data, etc.
    - adds complicating wrinkle to web scraping

Switch to Colab!

https://colab.research.google.com/drive/1lens27PQEW3-S2xYj776i190WpqNDISe#scrollTo=tIg6GIY3xODE&uniqifier=2

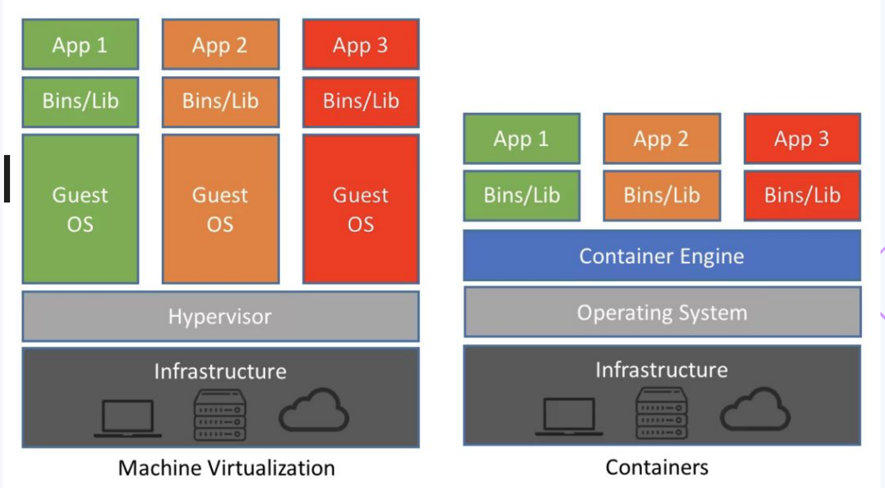
Docker and Containerization

Cloud Computing

- delivery of computing resources (hardware or software) as a service (owned and managed by the cloud provider)
- examples:
  - Maintaining a local GPU cluster vs. using Colab/AWS
  - Buying a physical hard disk for storage vs. using Google Drive/Dropbox
- **containers**: a popular method of enabling cloud-based app deployment

What is a container?

- Machine virtualization:
  - Run on top of the physical infrastructure
  - strong isolation but consumes resources
- Containers: share the host's OS kernel, use container engine to run applications
  - Lightweight, faster

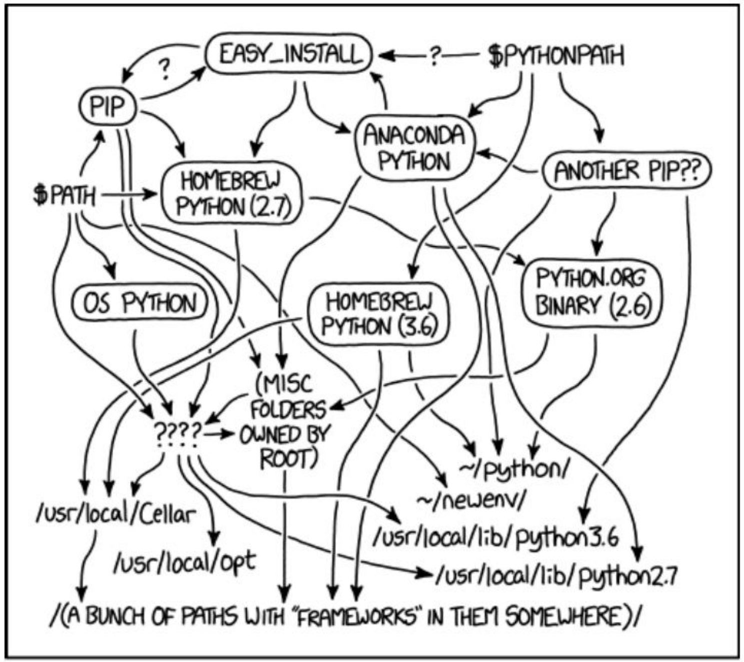


Source: Net App

Advantages of containerization

- highly portable:
  - all dependencies can be packaged into a single file
  - can be deployed to any server so long as the supporting container runtime is installed
- lightweight: don't have to bring an entire OS installation with the application
- reproducible: the host server does not care what happens inside the container and vice versa

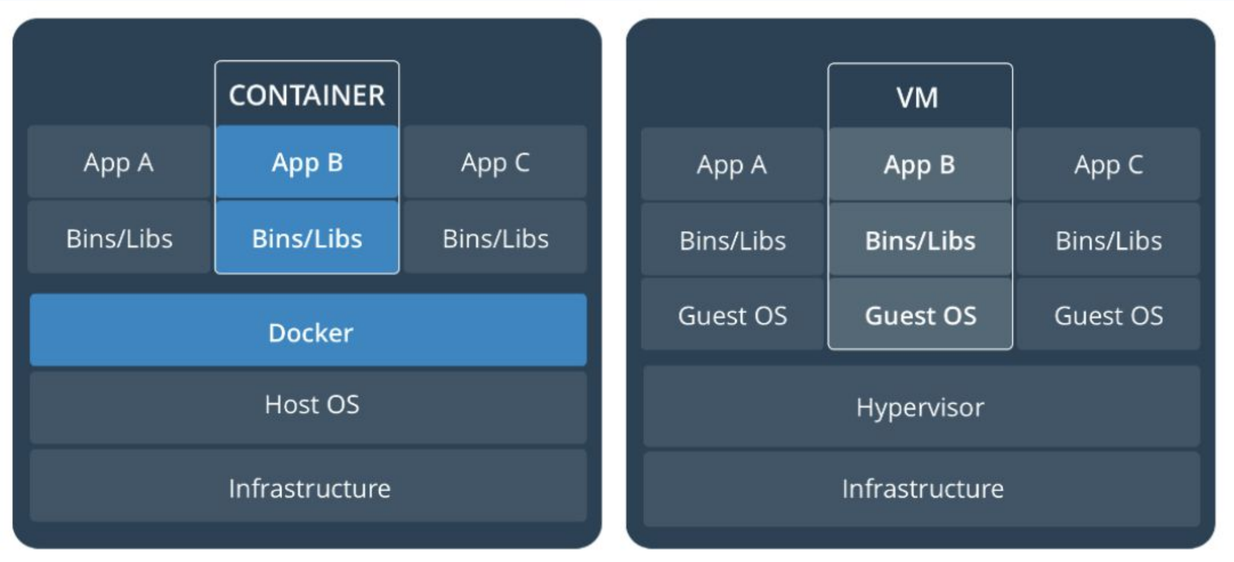
Advantages of containerization



MY PYTHON ENVIRONMENT HAS BECOME SO DEGRADED THAT MY LAPTOP HAS BEEN DECLARED A SUPERFUND SITE.

What is Docker?

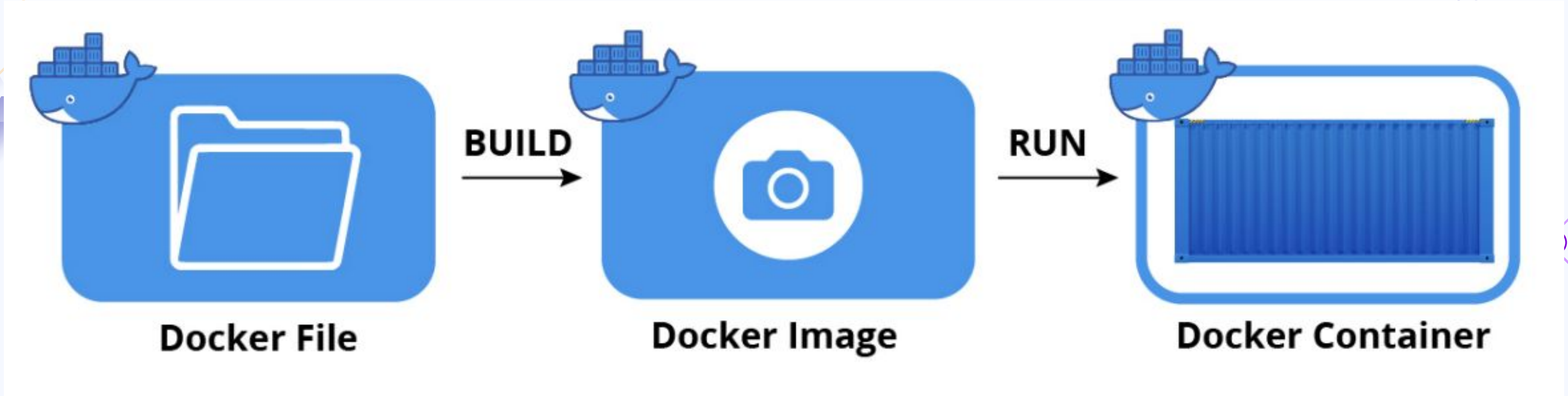
- a popular container ecosystem
  - for open source, see also: Podman



Docker Components

- Docker Engine: Container creation and management
- Docker Compose: Management of multi-container applications (e.g. separate containers for web server and database)
- Docker Hub: Repository for pulling/pushing Docker images
    - think Github but for containers

Docker container creation



Thanks!

Do you have any questions?

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